Homework 3

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```r
#Load the tidyverse package
library(tidyverse)

# Load the here package so file paths follow correctly
library(here)

#Read in the kelp data and store as an object called kelp
kelp <- read_csv(here("data", "temp-kelp.csv"))

#Read in personal data and store as an object called personal_data
personal_data <- read_csv(here("data", "personaldata2.csv"))
```

Problem 1. Giant kelp fronds

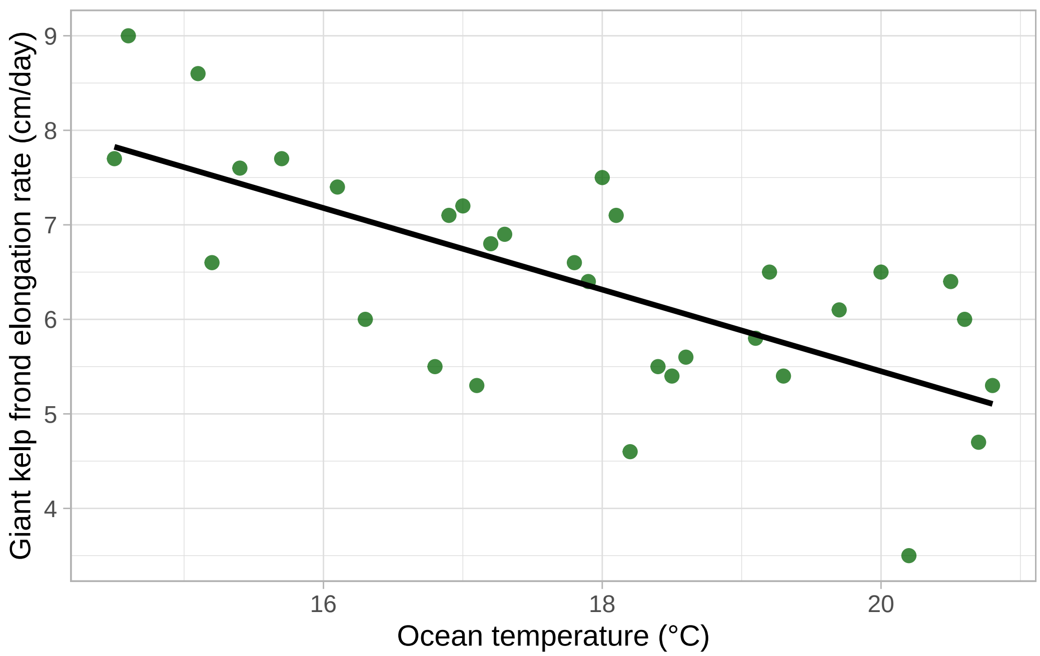
a. An appropriate test

In order to determine the strength of the relationship between temperature and giant kelp frond elongation rate, one may employ a Pearson correlation test, which measures how strong the relationship is and whether it is positive or negative. Additionally, one may utilize a simple linear regression test to determine whether ocean temperature predicts kelp elongation rate and what the growth rate is for each 1°C increase in temperature. Both tests are applicable because both variables measured are continuous.

b. Create a visualization

```r
#Create a scatterplot to display the relationship between temperature and
#giant kelp frond elongation rate.
ggplot(
  data = kelp,
  mapping = aes(x = temp_c,
                y = kelp_elong)) +

#Add data for each kelp observation in the data set and change color
geom_point(color = "darkgreen",
          size = 2,
          alpha = 0.75) +

#Add a linear trend line to decipher the relationship
geom_smooth(method = "lm",
```

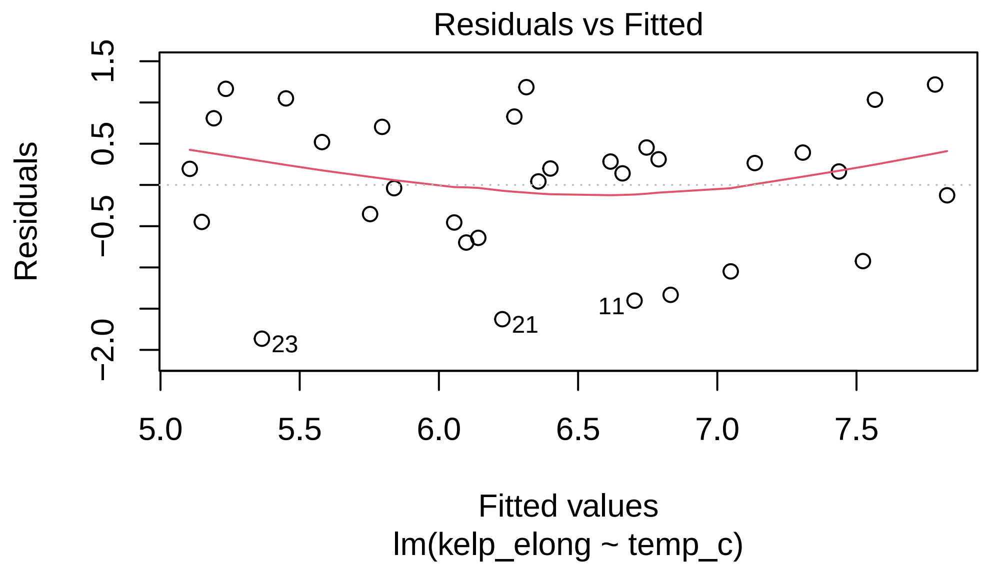
```
            se = FALSE,
            color = "black") +

#Relabel the both axes
labs(x = "Ocean temperature (°C)",
     y = "Giant kelp frond elongation rate (cm/day)") +

#Change theme
theme_light()
```

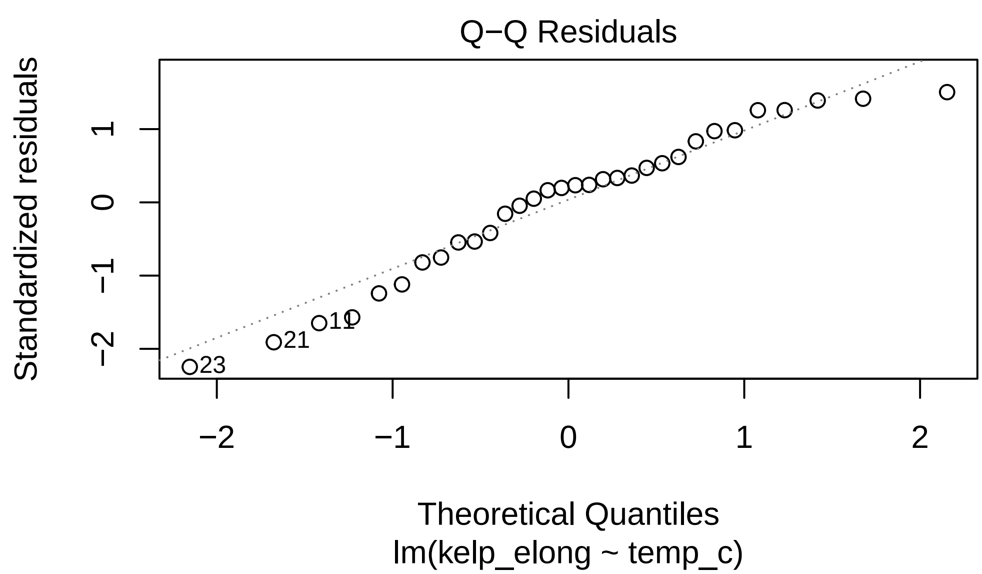


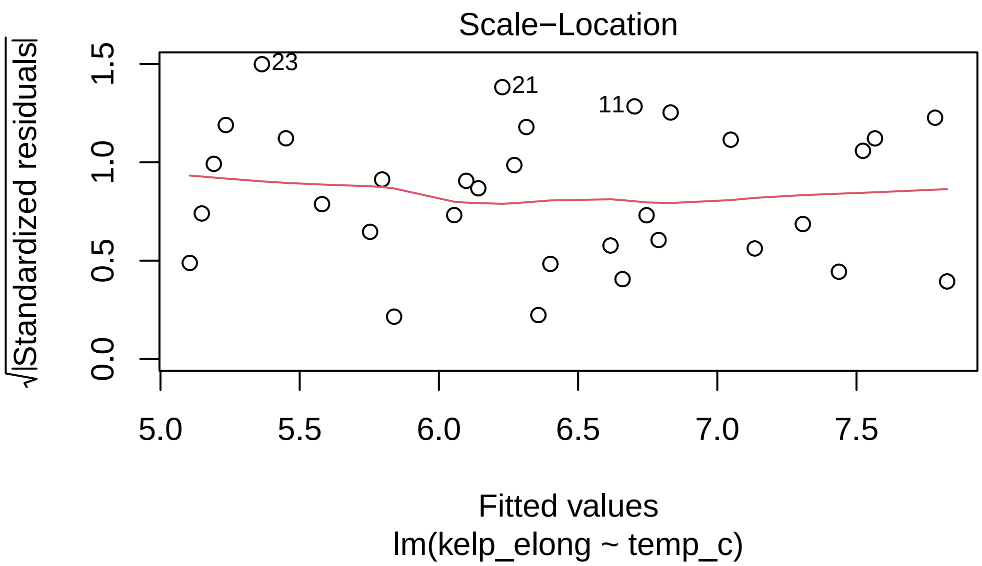
c. Check your assumptions and run your test

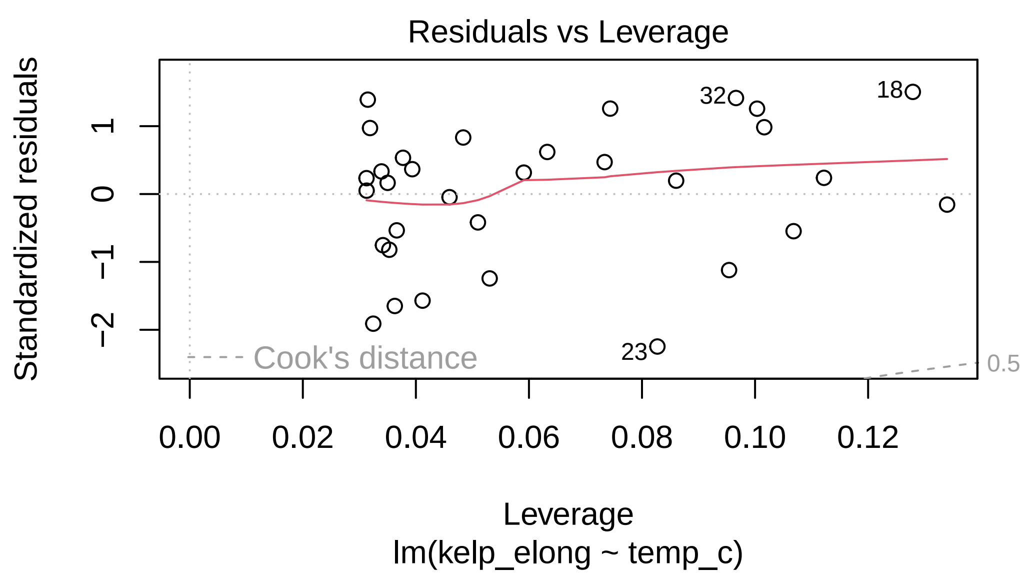
Check assumptions

```
#Create a linear regression model
kelp_lm <- lm(kelp_elong ~ temp_c,
              data = kelp)

#Create a diagnostic plots to check model assumptions
plot(kelp_lm)
```


Residuals vs Fitted
Residuals
1.5
0.5
−0.5
−2.0
23
21
11
5.0
5.5
6.0
6.5
7.0
7.5
Fitted values
lm(kelp_elong ~ temp_c)


Q−Q Residuals
Standardized residuals
1
0
−1
−2
23
21
11
−2
−1
0
1
2
Theoretical Quantiles
lm(kelp_elong ~ temp_c)





I checked the linearity assumption with the use of the scatterplot, which displayed a generally negative relationship between ocean temperature and giant kelp frond elongation rate. I then checked the assumptions using the diagnostic plots from the linear model, including the residuals vs. fitted plot, Q-Q plot, scale-location plot, and residuals vs. leverage plot. The residuals vs. fitted plot displayed a slight curve in the smoothed trend line, although there was no extreme pattern and the residuals were mostly spread around 0, so is reasonable to continue with a simple linear regression.

Run test

```r
#Summarize the linear regression model
summary(kelp_lm)
```

```

Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473,	Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

d. Results communication

In order to evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a simple linear regression, as kelp frond elongation rate is a continuous response variable and ocean temperature is a continuous predictor variable. The results suggest that ocean temperature significantly predicted kelp frond elongation rate (linear model: $F(1, 30) = 26.93$, $p < 0.001$, $\alpha = 0.05$, adjusted $R^2 = 0.455$). Therefore, for each 1°C increase in ocean temperature, one would expect a decrease in giant kelp frond elongation rate by 0.432 ± 0.083 (SE) cm/day.

e. Test implications

I would communicate to the team that higher ocean temperatures were generally associated with lower giant kelp frond elongation rates, according to this data set. As kelp elongation rate was shown to decrease by 0.432 cm/day for each 1°C increase in ocean temperature, areas with warmer ocean conditions may be prudent to monitor when looking at its relationship to kelp growth and overall kelp forest health. Despite this, as the study is based on one observational data set, it is important to communicate that this result shows an association rather than proving a causation.

f. Double check your own work

```r
#Run a Pearson correlation test to test the strength of the relationship
cor.test(x = kelp$temp_c,
         y = kelp$kelp_elong,
         method = "pearson")
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor 
-0.6877489 
```

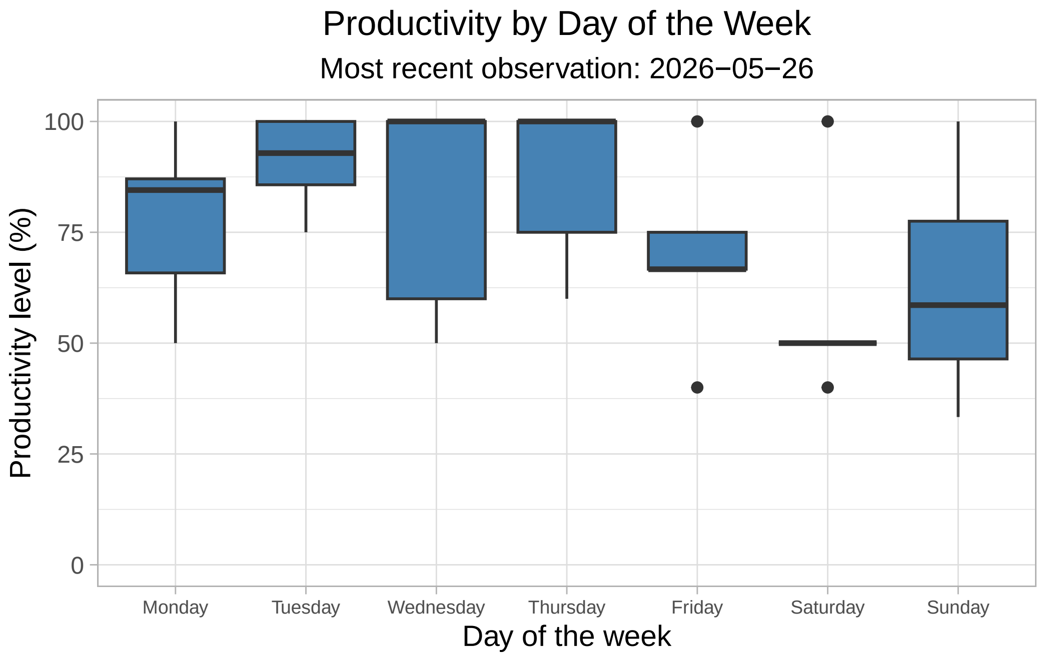
Both the Pearson correlation test and simple linear regression tests reject the null hypothesis, as both tests suggest a statistically significant negative relationship between ocean temperature and giant kelp frond elongation rate. The Pearson correlation test displays a moderately strong negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's r = -0.688, $t(30) = -5.19$, $p < 0.001$, $\alpha = 0.05$). This conclusion is in alignment with the results of the simple linear regression test, as they both suggest that kelp frond elongation rate decreased as ocean temperature increased.

Problem 2. Personal data

a. Updating your visualizations

Productivity by day of the week

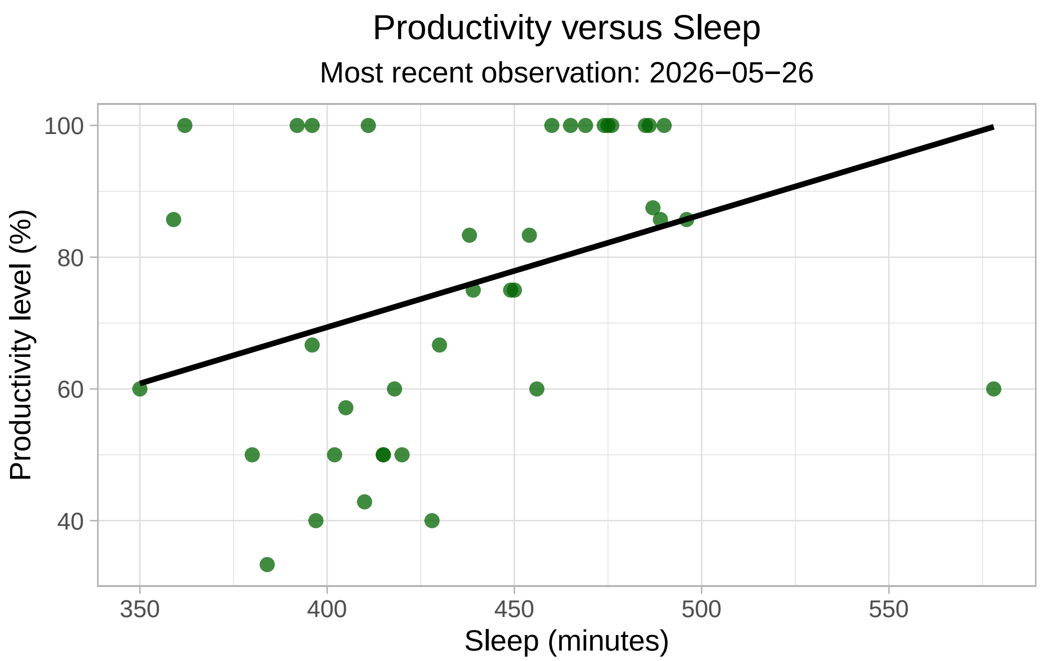
```r
#Define a productivity column as a percentage
personal_data <- personal_data |>
  mutate(productivity = tasks_completed / tasks_planned * 100)

#Create a boxplot to display productivity versus day of the week
ggplot(data = personal_data,
       mapping = aes(x = factor(day_of_week,
                                levels = c("Monday",
                                           "Tuesday",
                                           "Wednesday",
                                           "Thursday",
                                           "Friday",
                                           "Saturday",
                                           "Sunday")),
                     y = productivity)) +

#Set y-axis limits to show full range
scale_y_continuous(limits = c(0, 100)) +

#Add boxplots to summarize productivity depending on each day
geom_boxplot(fill = "steelblue") +

#Relabel both axes and add most recent observation date
labs(x = "Day of the week",
     y = "Productivity level (%)",
     title = "Productivity by Day of the Week",
     subtitle = "Most recent observation: 2026-05-26") +

#Change theme
theme_light() +

#Center title and subtitle and make x-axis smaller
theme(plot.title = element_text(hjust = 0.5),
      plot.subtitle = element_text(hjust = 0.5),
      axis.text.x = element_text(size = 7))
```



Productivity versus sleep

```
#Create a scatterplot to display relationship between sleep and productivity
#(tasks completed/tasks planned)
ggplot(data = personal_data,
       mapping = aes(x = sleep_minutes,
                     y = productivity)) +

#Add one point for each daily observation from data set
geom_point(color = "darkgreen",
           size = 2,
           alpha = 0.75) +

#Add a linear trend line to display relationship
geom_smooth(method = "lm",
            se = FALSE,
            color = "black") +

#Relabel both axes and insert most recent observation date
labs(x = "Sleep (minutes)",
     y = "Productivity level (%)",
```

```
     title = "Productivity versus Sleep",
     subtitle = "Most recent observation: 2026-05-26") +

#Change theme
theme_light() +

#Center the title and subtitle
theme(plot.title = element_text(hjust = 0.5),
      plot.subtitle = element_text(hjust = 0.5))
```



b. Captions

Figure 1. Productivity level varied by day of the week across the personal data collection period. Blue boxplots display the median and interquartile range of productivity level, measured as a percentage of tasks completed out of tasks planned at the beginning of the day, for each day of the week. Data were derived from personal observations collected from 2026-04-19 to 2026-05-26. Data accessed 2026-05-28.

Figure 2. Productivity level generally increased as sleep increased throughout the personal data collection period. Transparent dark green points represent daily observations of sleep, in minutes, and productivity level, measured as a percentage of tasks completed out of tasks planned at the beginning of the day. The black line shows the linear trend between sleep

and productivity. Data were derived from personal observations collected from 2026-04-19 to 2026-05-26. Data accessed 2026-05-28.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

For my affective visualization, I am thinking of creating a 3-dimensional model of a skyline, where exposure to light represents productivity level and the height of each building represents a range of minutes of sleep. I would orientate the buildings in a parallel line so the tallest building (representing most sleep) would be at the front, and closest to the light (representing the sun), meaning it would therefore be the most lit up. The subsequent buildings would then be partially covered by the building in front of them and receive less light, so the smallest building (representing the least sleep) would be the darkest and represent lower productivity. The angle of the trend line should follow the angle of descending buildings on the skyline as to properly explore the relationship between sleep and productivity level.

b. Create a sketch (on paper) of your idea.

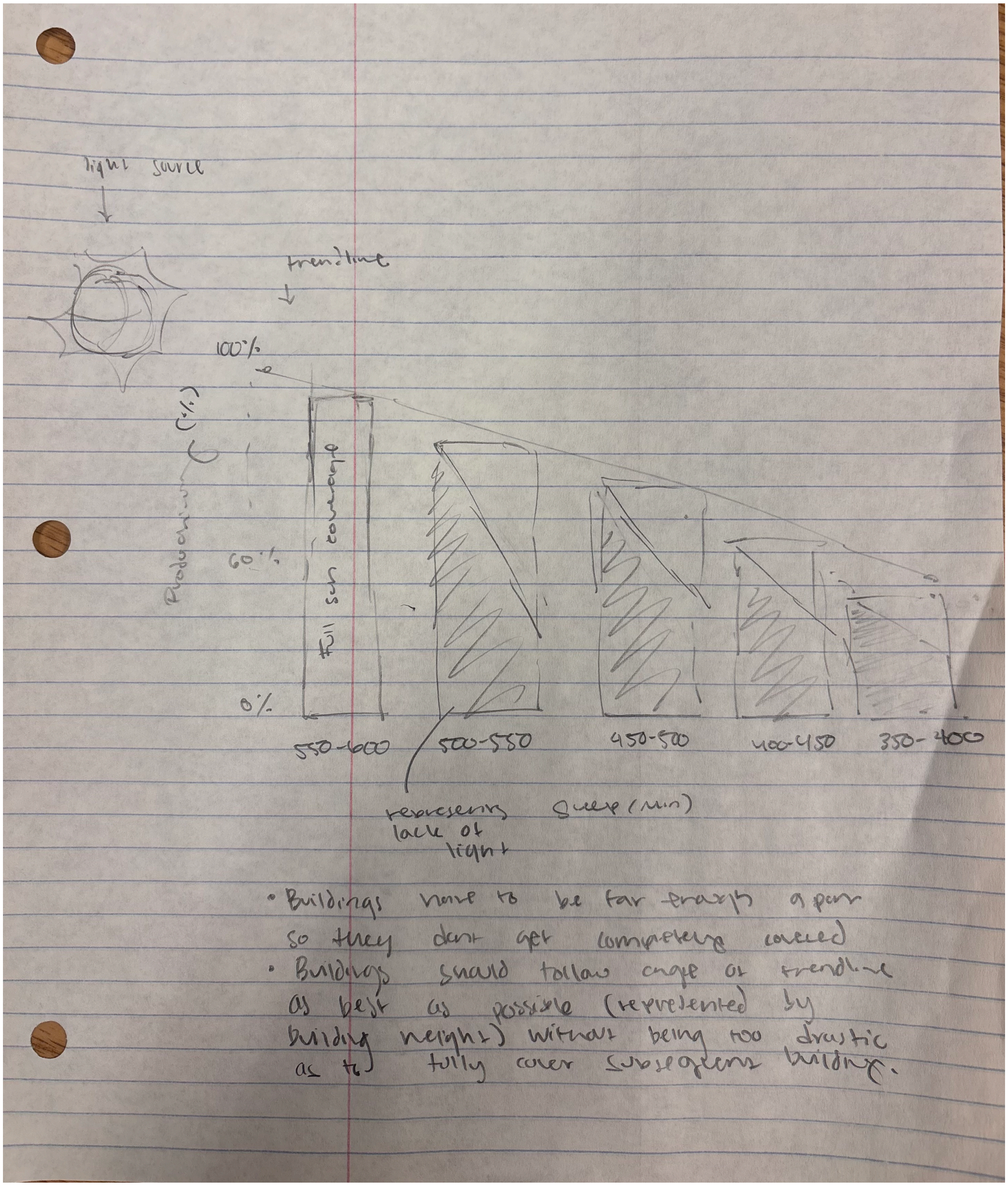


Figure 1: Sketch of affective visualization

c. Make a draft of your visualization

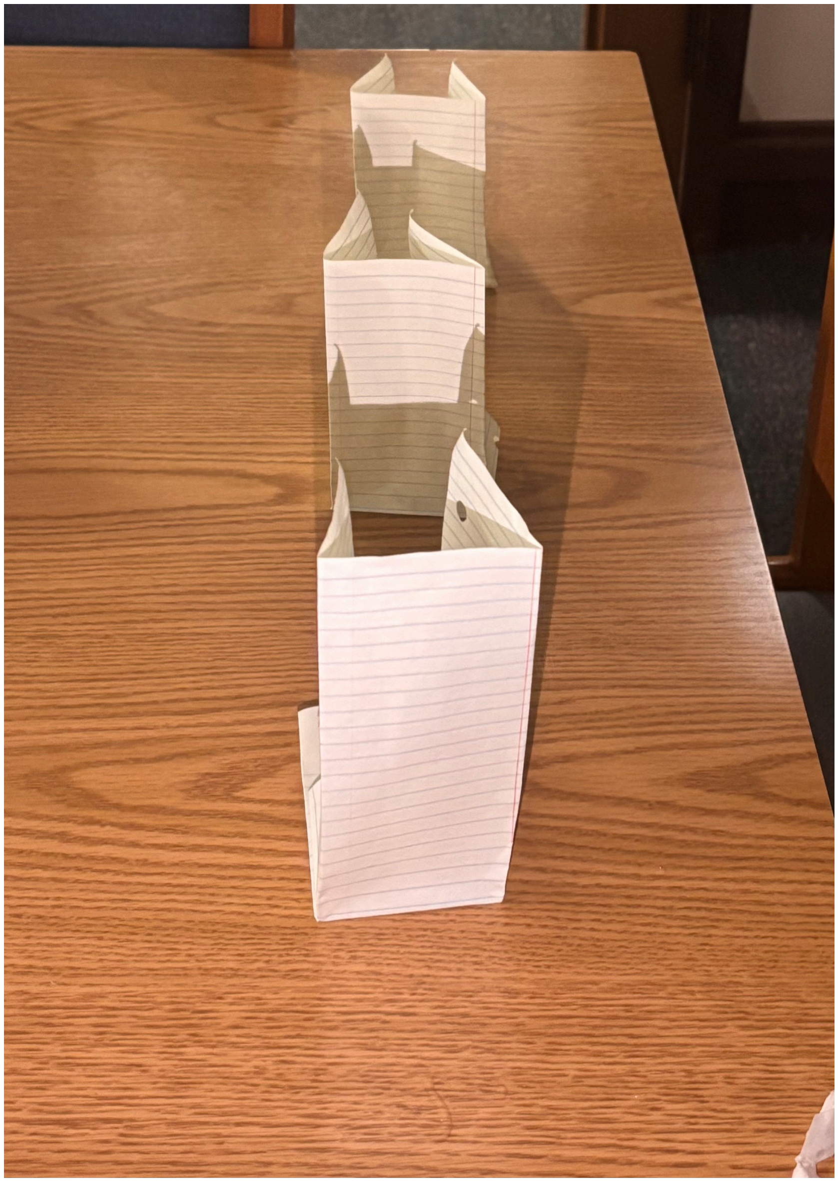


Figure 2: Draft of affective visualization

d. Write an artist statement

My affective visualization works to represent the relationship between sleep and productivity. I used a skyline of buildings to represent different amounts of sleep, with the taller buildings, that are exposed to more sunlight, representing more sleep. The shorter buildings are exposed to less sunlight and represent less sleep. By arranging the model to where the taller buildings receive more light, the model displays how sleep affects ones ability to complete more tasks, and therefore their level of productivity.

e. Prep your materials to share in class

Slides for affective visualization

Problem 4. Statistical critique

a. Revisit and summarize

I chose to look at the linear regression test performed in this experiment, which looks at the relationship between that of the pH levels in the stream and its relationship to the streams biological health, based on the IBI (Index of Biolocal Integrity) model values. The response variable is the IBI scores, or how biologically "healthy" the system is, and the predictor variable, in this case, is the sample pH of the stream.

Dependent variable	**Independent variable**	R^2	***F*-statistics**	***d.f.***	***p*-Values**
IBI scores (n=5)	Water temperatures	0.005	0.002	3	0.971
	Dissolved oxygen	0.270	1.108	3	0.370
	pH	0.785	10.953	3	0.045*
	Turbidity	0.600	4.497	3	0.124
	Conductivity	0.486	2.839	3	0.190

R^2: regression coefficients; *d.f.*: degrees of freedom.

*

p=0.05.

Figure 3: Table described in Homework 2

b. Visual clarity

The table clearly represents the summary statistics from the regression tests as each independent variable is listed in its own row and the statistical outputs are separated into columns. This makes it easy to compare which physico-chemical variables were more strongly related to IBI scores, especially because pH has the highest R^2 value and is marked as statistically significant with an asterisk.

c. Aesthetic clarity

The table has relatively low visual clutter as it uses a simple layout with clear rows and columns. Ultimately, most of the table is used to present statistical information rather than unnecessary decoration.

d. Recommendations

I would recommend bolding or visually emphasizing the pH row because as it is the only statistically significant result in the table. Furthermore, I would also recommend adding a scatterplot showing pH on the x-axis and IBI score on the y-axis. This would ultimately make the relationship easier to interpret as the table asserts a statistical summary, but fails to show the underlying observations and data.